Engineering Mathematics: Complex Variable, second mock midterm question sheet. 工程數學：複變期中考第二次模擬考題目卷。The exam runs from 10:30 to 12:00. 考試時間 10:30–12:00。Printed materials are permitted during the final hour. 最後一小時開放使用紙本資料。Internet and AI tools (on phones, tablets, or laptops) are permitted during the final ten minutes. 最後十分鐘開放使用網路與 AI（手機、平板、筆電皆可使用）。Do not get help from real people (especially your classmates), so messaging apps, social networks, shared Google Docs, and similar features are not allowed, and neither is a shared AI subscription account. 請不要從真人（尤其是你的同學）得到幫助，所以通訊軟體、社群網站、共享 Google Docs、類似的功能都不能用，共用 AI 訂閱帳號也不行。Write your answers on the answer sheet; the question sheet will not be collected. 答案寫在答案卷上，題目卷不回收。Unless otherwise stated, each problem is worth one point, only completely correct answers receive credit; wrong answers are not penalized. 除非特別註明，每題一分，全對才給分，答錯不倒扣。

❶ Write your name. 寫下你的姓名。 ❷ And your student ID. 以及你的學號。 ❸ Remove all letters from your student ID and read the remaining digits as a decimal number. Multiply it by 1002003 and reduce modulo 10^8, that is, keep only the last eight digits. Write the result as eight digits, padding with leading zeros if necessary, and call them $q_1 q_2 \cdots q_8$ from left to right; for example, B12345678 gives $12345678 \times 1002003 = 12370406393034$, whose last eight digits are 06393034, so $q_1 = 0$, $q_2 = 6$, …, $q_8 = 4$. Later problems use these digits. If a mistake here carries into later problems, you lose only this point; the later problems are graded using the digits you write here. 把學號中的字母全部去掉，剩下的數字當作一個十進制數。將它乘以 1002003 之後只留最後八位。把結果寫成八位數，不足八位時往前補零，由左到右記作 $q_1 q_2 \cdots q_8$；例如 B12345678 得到 $12345678 \times 1002003 = 12370406393034$，最後八位是 06393034，所以 $q_1 = 0$、$q_2 = 6$、……、$q_8 = 4$。後面的題目會用到這些數字。如果你在這題算錯導致後面跟著錯，你只會損失這題的一分；後面的題目會用你在這裡寫的數字重新計算答案。

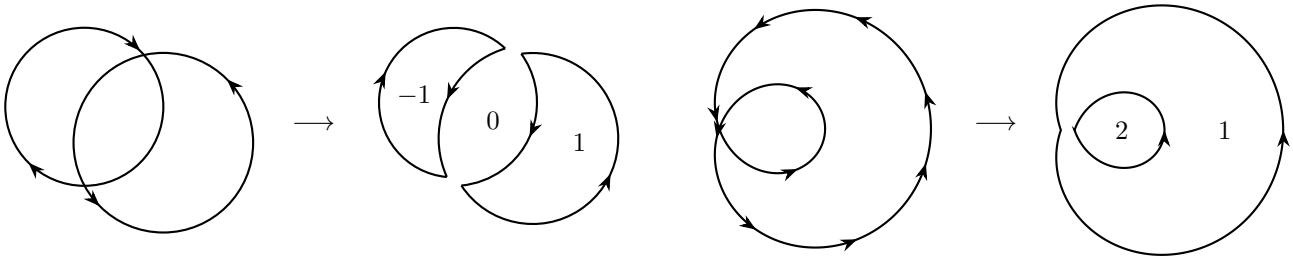
● True or false? For question $10 + j$, answer q_j if the statement is true, and answer $9 - q_j$ if the statement is false. In this group, Log and Arg have their branch cut along the parabola $y = -x^2$, $x \le 0$, and $\operatorname{Arg} 1 = 0$; this means that Log and Arg are continuous at every point off this parabola. 是非題，如果第 $10 + j$ 題的敘述為真，回答 q_j，否則回答 $9 - q_j$。這組題目中 Log 與 Arg 的 branch cut 在拋物線 $y = -x^2$，$x \le 0$ 上，且 $\operatorname{Arg} 1 = 0$；這表示 Log 與 Arg 在這條拋物線以外的每一點都連續。 ⓫ Log is continuous at -1. ⓬ $\operatorname{Arg}\big((1+i)^{q_8 - q_3}\big) = (q_8 - q_3)\pi/4$. ⓭ $\operatorname{Arg}(zw) = \operatorname{Arg} z + \operatorname{Log} w$ whenever z and w are real numbers. ⓮ Off the parabola, $\sqrt{z}$ is well-defined and differentiable as $\exp(\operatorname{Log} z/2)$. ⓯ $\{z \in \mathbb{C} : |z| \ge 1\}$ is a closed subset of $\mathbb{C}$. ⓰ $\oint_{|z|=1} -\bar{z}^{-n}\, dz = 0$ for all integers $n \ge 1$. ⓱ Two functions holomorphic on $\mathbb{C} \setminus \{\infty\}$ that agree at $z = 1/n$ for every integer $n \ge 1$ are equal. ⓲ If $|c_n| \to 0$ but $\sum |c_n|$ diverges, then for every $t \in \mathbb{C}$ some rearrangement of $\sum c_n$ converges to t.

● Find each integral; every circle is traversed counterclockwise. 求下列積分；每個圓都是逆時針方向。 ⓳ $\oint_{|z-2|=1} z/(z^2 - 4)\, dz$. ⓴ $\oint_{|z|=1} z^{-3} e^{z^2}\, dz$. ㉑ $\oint_{|z|=1} e^z/z^2(z-2)\, dz$. ㉒ $\oint_{|z|=1} z/(1 - \cos z)\, dz$. ㉓ $\oint_{|z-2\pi i|=1} dz/(e^z - 1)$. ㉔ $\oint_C \bar{z}\, dz$, where C is the boundary of the triangle with vertices 0, $q_4 + 1$, and $q_4 + 1 + (q_1 + 1)i$, traversed counterclockwise.

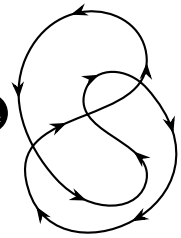
● Answer C if the series converges and D if it diverges. 級數收斂回答 C，發散回答 D。 ㉕ $\sum_{n \ge 1} (-1)^n/\sqrt{n}$. ㉖ $\sum_{n \ge 1} (1 - 1/n)^n$. ㉗ $\sum_{n \ge 0} \big((1 + 2i)/3\big)^n$.

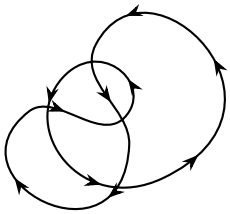
● YangMing writes each function below in terms of x, y, $z = x + iy$, and $\bar{z} = x - iy$; each is defined wherever the formula makes sense. Answer Y if it satisfies Cauchy–Riemann equations at all places that the function is defined. ㉘ $\bar{z} + \big(1 + (-1)^{q_2}\big) iy$. ㉙ $x^3 - 3xy^2 - x + i(3x^2 y - y^3 - y)$. ㉚ $z^2 + \bar{z}^2$. ㉛ $e^{\bar{z}}(\cos 2y + i \sin 2y)$. ㉜ $z\bar{z} + 2(-1)^{q_1} y^2 + 2ixy$. ㉝ $x^2 - y^2 + x + i(2xy - y)$.

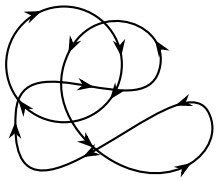
● Throughout the course, we have shown that two contour integrals are equal by recombining the contours into several disjoint loops so that Cauchy's theorem can be applied to each loop. 這學期我們多次把圍道重新組合成幾個互不相交的 loops，對每個 loop 使用 Cauchy's theorem，來證明兩個圍道積分相等。

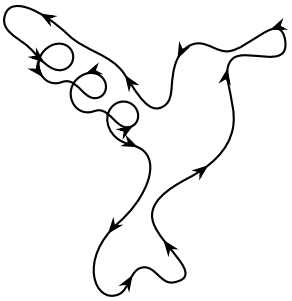


Do the same for the figures below. 對下面的圖做同樣的事。

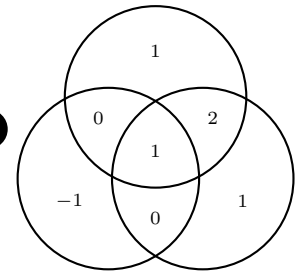
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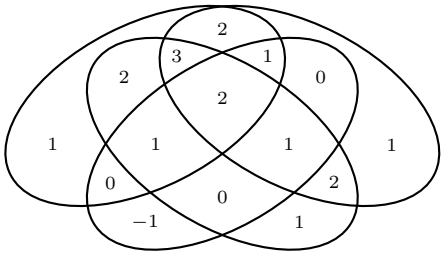
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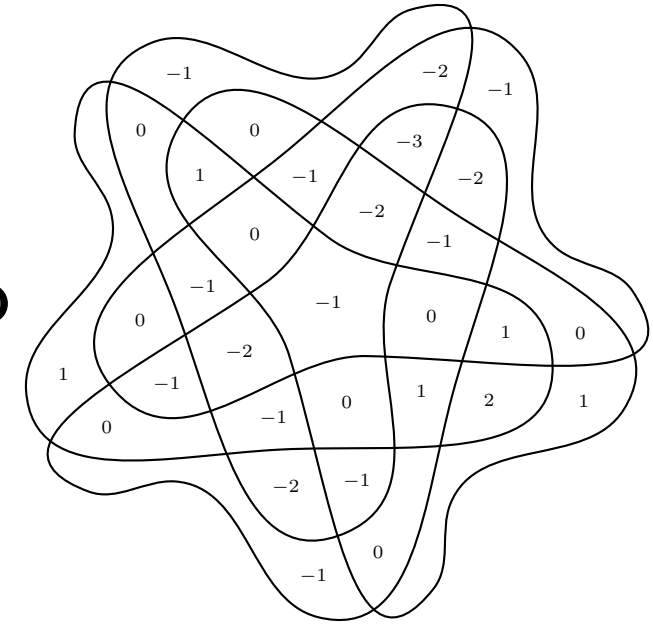
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● Now the other way around. You are given the winding numbers of each region, and you need to mark the direction of each closed curve.

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● TaTung computes Schwarzian derivatives: for f with $f' \neq 0$, $S(f) = (f''/f')' - (f''/f')^2/2$. TaTung 計算 Schwarzian derivative：對 $f' \neq 0$ 的 f，$S(f) = (f''/f')' - (f''/f')^2/2$。 41 $S(f)$ for $f(z) = e^{(q_2+1)z}$. 求 $f(z) = e^{(q_2+1)z}$ 的 $S(f)$。 42 $S(f)$ for $f(z) = z^{q_7+2}$, $z \neq 0$. 求 $f(z) = z^{q_7+2}$（$z \neq 0$）的 $S(f)$。 43 $S(f)$ for $f(z) = (2z+1)/(z+3)$. 求 $f(z) = (2z+1)/(z+3)$ 的 $S(f)$。 44 $S(f)$ for $f(z) = \tan z$. 求 $f(z) = \tan z$ 的 $S(f)$。

● DongHwa continues a function analytically by re-expanding power series. Let h be holomorphic on $\mathbb{C} \setminus \mathbb{Z}$, and let $h(z) = \sum_{n\geq 0} a_n(z-i)^n$ be its Taylor series about i; by Taylor's theorem it converges on the largest open disk centered at i that contains no integer. Each jump below moves the center to a point inside the current disk of convergence and re-expands h there. DongHwa 用重新展開冪級數的方法做 analytic continuation。設 h 在 $\mathbb{C} \setminus \mathbb{Z}$ 上 holomorphic，$h(z) = \sum_{n\geq 0} a_n(z-i)^n$ 是它在 i 的 Taylor series；由 Taylor's theorem，它在以 i 為圓心、不含任何整數的最大開圓盤上收斂。下面每一次「跳」都是把圓心移到目前收斂圓盤內的一點，再把 h 在那裡重新展開。 45 Can DongHwa get from i to $1000 + i$ in at most 1010 jumps? Answer Y or N. DongHwa 能不能在 1010 次以內從 i 跳到 $1000 + i$？回答 Y 或 N。

● FengChia studies where the roots of p' can be. Let $p(z) = c(z - z_1)(z - z_2)\cdots(z - z_n)$ with $c \neq 0$ and $n \geq 2$. FengChia 研究 p' 的根可能在哪裡。設 $p(z) = c(z - z_1)(z - z_2)\cdots(z - z_n)$，其中 $c \neq 0$、$n \geq 2$。 46 Write $p'(z)/p(z)$ as a sum over the roots z_k. 把 $p'(z)/p(z)$ 寫成對所有根 z_k 的和。 47 Let w be a root of p' that is not a root of p. Take the complex conjugate of $p'(w)/p(w) = 0$, and find positive numbers t_k with $\sum_k t_k(w - z_k) = 0$. 設 w 是 p' 的根但不是 p 的根。對 $p'(w)/p(w) = 0$ 取共軛，找出正數 t_k 使得 $\sum_k t_k(w - z_k) = 0$。 48 Write w as a weighted average of the z_k, with nonnegative weights adding up to 1. 把 w 寫成 z_k 的加權平均，權重非負且和為 1。

● TaiPei uses Hadamard's three-circle theorem: if f is holomorphic on the closed annulus $r_1 \leq |z| \leq r_3$ and $M(r)$ denotes the maximum of $|f|$ on the circle $|z| = r$, then for $r_1 \leq r_2 \leq r_3$,

$$\log M(r_2) \leq \frac{\log(r_3/r_2)\log M(r_1) + \log(r_2/r_1)\log M(r_3)}{\log(r_3/r_1)}.$$

In each problem, f is holomorphic on the closed annulus between the two given circles. TaiPei 使用 Hadamard's three-circle theorem：若 f 在閉環形區域 $r_1 \leq |z| \leq r_3$ 上 holomorphic，$M(r)$ 是 $|f|$ 在圓 $|z| = r$ 上的最大值，則對 $r_1 \leq r_2 \leq r_3$ 上式成立。每一題中的 f 都在兩個給定的圓之間的閉環形區域上 holomorphic。 49 Suppose $|f| \leq 1$ on $|z| = 1$ and $|f| \leq 4^{q_8+1}$ on $|z| = 4$. What upper bound for $|f|$ on $|z| = 2$ does the theorem give? 設在 $|z| = 1$ 上 $|f| \leq 1$，在 $|z| = 4$ 上 $|f| \leq 4^{q_8+1}$。這個定理給出 $|f|$ 在 $|z| = 2$ 上的什麼上界？ 50 Suppose $|f| \leq 8$ on $|z| = 1$ and $|f| \leq 1$ on $|z| = 8$. What upper bound for $|f|$ on $|z| = 2$ does the theorem give? 設在 $|z| = 1$ 上 $|f| \leq 8$，在 $|z| = 8$ 上 $|f| \leq 1$。這個定理給出 $|f|$ 在 $|z| = 2$ 上的什麼上界？ 51 For $f(z) = z^n$ with an integer n, does equality hold in the theorem for all $r_1 \leq r_2 \leq r_3$? Answer Y or N. 對 $f(z) = z^n$（n 為整數），定理中的不等式是否對所有 $r_1 \leq r_2 \leq r_3$ 都取等號？回答 Y 或 N。

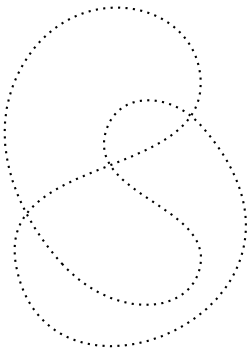
● ShihChien computes $\sum_{n\geq 1} n^{-2}$. Let $g(z) = \pi\cot(\pi z)z^{-2}$, and for $N = 1, 2, \ldots$ let C_N be the boundary of the square with vertices $(N + 1/2)(\pm 1 \pm i)$, traversed counterclockwise. You may use that $|\cot(\pi z)| \leq 2$ on every C_N. ShihChien 要計算 $\sum_{n\geq 1} n^{-2}$。令 $g(z) = \pi\cot(\pi z)z^{-2}$；對 $N = 1, 2, \ldots$，令 C_N 是頂點為 $(N + 1/2)(\pm 1 \pm i)$ 的正方形的邊界，逆時針方向。可以直接使用：在每個 C_N 上 $|\cot(\pi z)| \leq 2$。 52 Find the coefficient of z^{-1} in the Laurent series of g about 0. 求 g 在 0 的 Laurent series 中 z^{-1} 的係數。 53 For an integer $n \neq 0$, find the coefficient of $(z - n)^{-1}$ in the Laurent series of g about n. 對整數 $n \neq 0$，求 g 在 n 的 Laurent series 中 $(z - n)^{-1}$ 的係數。 54 What upper bound for $\left|\oint_{C_N} g(z)\,dz\right|$ does the ML-inequality give? ML-inequality 給出 $\left|\oint_{C_N} g(z)\,dz\right|$ 的什麼上界？ 55 The integral $\oint_{C_N} g(z)\,dz$ is $2\pi i$ times the sum, over all poles of g inside C_N, of the coefficients found in the previous two problems. Let $N \to \infty$ to find $\sum_{n\geq 1} n^{-2}$. 積分 $\oint_{C_N} g(z)\,dz$ 等於 $2\pi i$ 乘以前兩題的係數對 C_N 內所有 poles 的和。令 $N \to \infty$，求 $\sum_{n\geq 1} n^{-2}$。

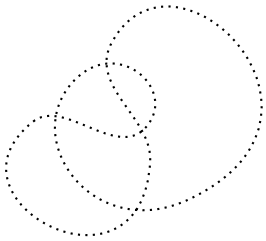
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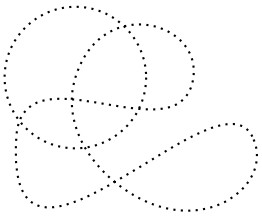
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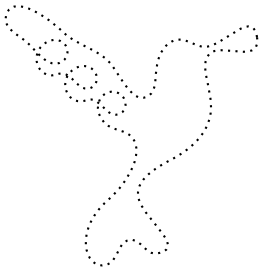
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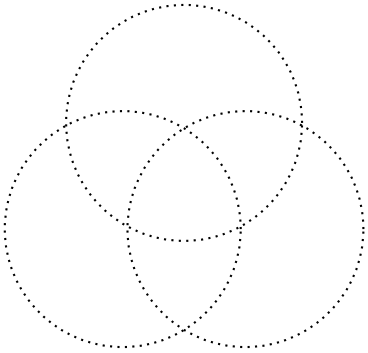
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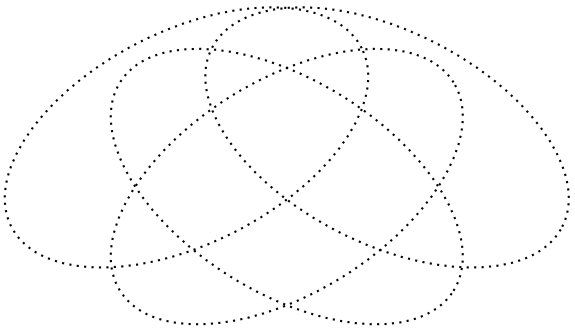
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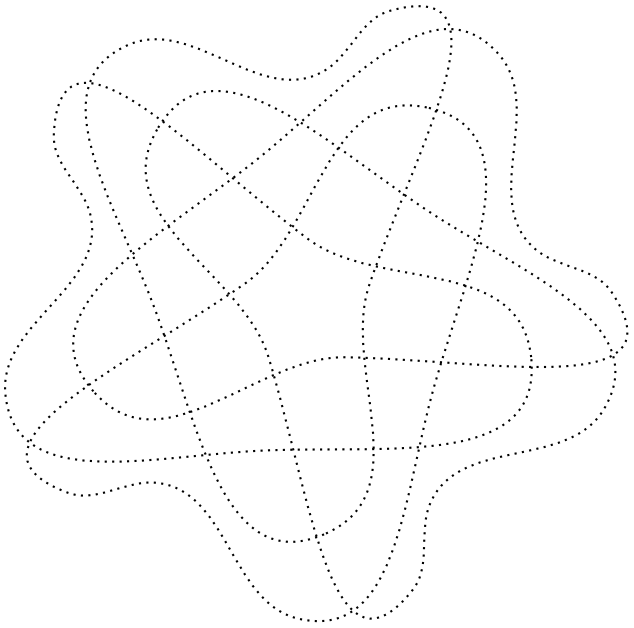
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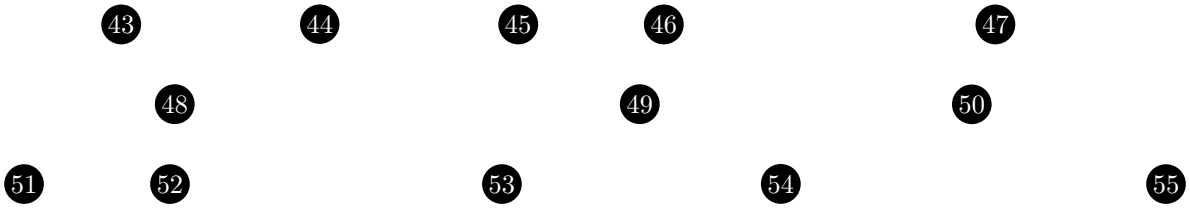
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